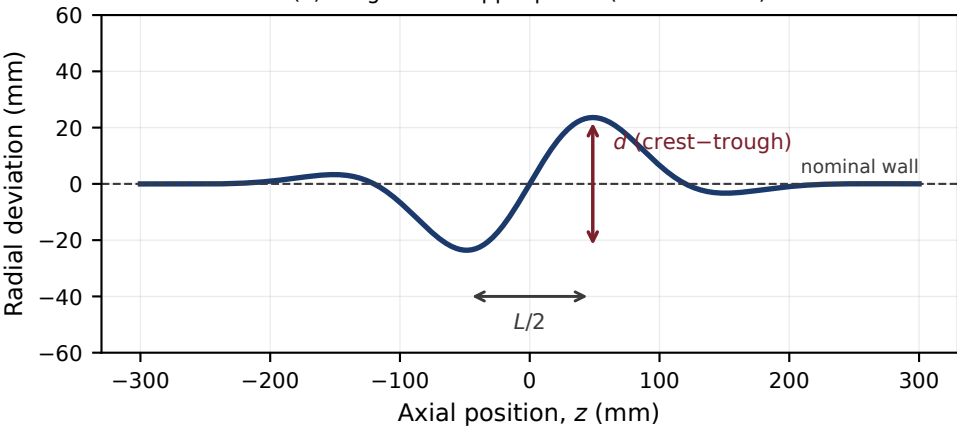
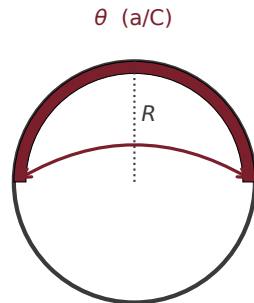


(a) Longitudinal ripple profile (raised-cosine)



(b) Circumferential extent



$D/t = 73.1$ (constant)
API 5L X70
 $P = 9.55$ MPa (MAOP)
 $a/C = 0.25 / 0.375 / 0.50$
 $\theta = 90^\circ / 135^\circ / 180^\circ$